HIP: Hessian Interatomic Potentials without derivatives

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Abstract

Molecular Hessians, the second derivatives of the potential energy, are fundamental to many workflows in computational chemistry. Usually, accurate Hessians are computationally expensive to calculate and scale poorly with system size, whether computed using quantum chemistry methods or machine-learning interatomic potentials (MLIPs). In this work, we introduce Hessian interatomic potentials (HIPs), a deep learning model that directly predicts Hessians without relying on automatic differentiation or finite differences. To do so, we construct SE(3)-equivariant, symmetric Hessians from irreducible representation (irrep) features up to degree l=2, computed by a graph neural network. HIP Hessians are one to two orders of magnitude faster, more accurate, more memory efficient, easier to train, and exhibit favourable scaling with system size. We validate our predictions across a wide range of downstream tasks, demonstrating consistently superior performance in transition state search, geometry optimization, zero-point energy corrections, and vibrational analysis. We open-source the HIP code and model weights.

1 Introduction

Molecular simulation is a cornerstone of material discovery and molecular design. While the prediction of energy and forces is often sufficient for structural dynamics, advanced computational chemistry workflows often depend on the molecular Hessian[1, 2]. By describing the local curvature of the potential energy surface (PES), the Hessian is foundational to many computational workflows. Second-order geometry optimization relies on the curvature information to rapidly and reliably converge to equilibrium structures[3]. Transition state (TS) searches and extrema classification depend on the Hessian to correctly identify saddle points and reaction barriers, which are essential for parameterizing reaction rates and microkinetic models for longer time and length scale simulations[4]. Furthermore, mode-following transition state search methods directly leverage the curvature information to locate the transition states[5]. Finally, vibrational analysis uses Hessian eigenvalues to compute zero-point energy (ZPE) corrections and thermodynamic free energies bridging the gap from static molecules to non-zero temperature observables[6].

For a molecule with N atoms, the Hessian is a $3N \times 3N$ matrix, where each entry requires a mixed second-order derivative of the electronic energy with respect to two nuclear coordinates. Despite their broad utility, Hessian calculations represent a substantial computational bottleneck. Traditional approaches to obtain the Hessian rely either on analytic second derivatives or on numerical differentiation of gradients, both of which are very expensive, and analytical Hessians in particular are often so complicated to implement that they are not widely supported for all methods [7–9].

Most computational chemistry calculations are carried out with Kohn-Sham density functional theory [10] (DFT), due to its balance of computational cost and accuracy [11]. However, the $O(N^4)$ scaling of hybrid DFT with system size remains expensive, which has spurred the development of machine-learning interatomic potentials (MLIPs). MLIPs promise to deliver near-DFT accuracy at a fraction of the cost while incorporating symmetries, thereby drastically reducing the need for extensive training data [12–17]. While recent work demonstrates that automatic differentiation (AD) can successfully recover the Hessian [18–23], this approach faces two substantial limitations. High accuracy of the energy and forces does not guarantee comparably accurate second derivatives. Much like training on force labels is needed to achieve accurate force predictions [24], explicit Hessian training labels are required for reliable AD Hessians [23, 25]. Second, these AD Hessians introduce high computational costs, with an extra factor of $\sim 3N$ during training and inference [19–23]. Rodriguez et al. [21] in particular showed that including the Hessian in the loss improves data efficiency and extrapolation of energy and force predictions but is limited by approximately 25 times longer training times, which make AD Hessians prohibitive to train. The same principle extends to learned exchange-correlation functionals, where supervising energy derivatives substantially improves accuracy [26]. MLIPs can also be used to obtain Hessians with finite differences [20, 27], an approach which requires less memory but $6N$ forward passes and also requires specialized data collection schemes to be accurate [28].

In this work, we introduce **HIP**: **H**essian **I**nteratomic **P**otentials. HIPs predict the Hessian directly, to eliminate the need for finite differences, coupled-perturbed equation solvers, or automatic differentiation. By building on SE(3)-equivariant neural networks, HIP constructs the full $3N \times 3N$ matrix Hessian at a fraction of the cost of traditional methods. We demonstrate that HIP accelerates predictions 10-70$\times$ and reduces peak memory by a factor of 2-3 compared to AD, while maintaining a highly favourable $O(N)$ scaling with the number of atoms. Furthermore, HIP achieves superior predictive accuracy, halving the mean absolute error compared to AD. We validate the practical utility of HIP across several critical downstream applications, demonstrating consistently superior performance in second-order geometry optimization, ZPE corrections, TS searches and frequency analysis for extrema classification.

2 Results

2.1 HIP framework for Hessian prediction

Equivariant graph neural networks (GNNs) efficiently predict scalar energies and forces either by direct equivariant prediction or conservatively through AD of the energy.[12, 29–31]. Computing the Hessian via AD incurs approximately $3N$ times the computational cost of a regular forward pass. To avoid this cost, we introduce a Hessian readout head that constructs the Hessian matrix directly in parallel (Fig. 1).

Our approach decomposes the Hessian into 3×3 equivariant sub-blocks $\mathbf{H}_{IJ}$, where I and J index atoms. Instead of standard message aggregation, we use node features to compute self-interactions ($\mathbf{H}_{II}$) and treat message-passing features between atoms I and J as direct predictions of pair interactions ($\mathbf{H}_{IJ}$). Using Clebsch–Gordan tensor products, we map irreducible representations that include angular momentum degree $l = 2$ into Cartesian 3×3 matrices. This basis change guarantees SE(3) equivariance and structural symmetry by design.

This readout head can be appended to any equivariant backbone, such as EquiformerV2[32] used in this work, while retaining the base model's $O(N)$ global scaling. The Hessian interaction cutoff can also be set independently of the backbone radius to represent the locality of quantum-mechanical interactions.

2.2 HIP Hessian accuracy

2.2.1 Benchmark

We assess Hessian quality using the element-wise mean absolute error (MAE), the MAE of the eigenvalues λ and the smallest eigenvalue λ_1, and the cosine similarity of the smallest eigenvector v_1. We focus particularly on the lowest eigenvalue and eigenvector as they play an important role in downstream applications such as transition state search. We also measure the average inference time. Table 1 shows results on the Hessian dataset for Optimizing Reactive MLIP (HORM)-T1x validation set. Models trained only on energy and force labels, such as EquiformerV2 (E-F), have substantially higher Hessian errors, showing that accurate second derivatives require explicit Hessian training. Among the methods using automatic differentiation with Hessian training data, EquiformerV2 has the lowest errors. HIP-EquiformerV2 improves on all baselines across every reported metric. Compared with AD-EquiformerV2, HIP has 2.5× lower Hessian MAE, 3.8× lower eigenvalue MAE and higher eigenvector cosine similarity, while being 16× faster. End-to-end HIP training (marked by * in Table 1) reduces the errors further, reaching a Hessian MAE of 0.020 eV/Å^2 at 31.4 ms per prediction.

2.2.2 Glycine proton transfer

The aggregate errors in Table 1 summarize Hessian quality as scalar averages. To probe the curvature along a physically meaningful example, we study the intramolecular proton transfer of glycine from the Transition1x test split [33]. The motion is described by the N–H and O–H distances of the transferring proton, q_{NH} and q_{OH}, which we transform to the reaction coordinate $\xi = q_{\mathrm{NH}} - q_{\mathrm{OH}}$ (Methods, Glycine proton-transfer case study).

We study the two-dimensional $(q_{\mathrm{NH}}, q_{\mathrm{OH}})$ surface around the proton-transfer transition state. At each grid point, the structure is relaxed at the DFT level while q_{NH} and q_{OH} are constrained to their target values. We then characterize the surface by counting the negative Hessian eigenvalues (Fig. 2a). HIP reproduces the DFT topology closely, while AD introduces spurious negative modes in the transition-state and minimum regions.

We then track the six lowest vibrational eigenvalues along the all-atom minimum-energy path as a function of the reaction coordinate ξ (Fig. 2b). HIP overlaps the DFT reference across all six modes. AD Hessians capture only the lowest mode but systematically underestimate the orthogonal curvatures of modes 2–6. This view matches the gap in Table 1, where HIP captures the full curvature spectrum that AD Hessians miss.

Finally, we project the Cartesian force $\mathbf{F}$ onto the unit tangent $\widehat{\boldsymbol{\xi}}$ of the minimum-energy path, $\mathbf{F} \cdot \widehat{\boldsymbol{\xi}}$ (Fig. 2c). We show EquiformerV2 (EqV2) and LeftNet with conservative forces (LeftNet-CF), each differentiated to obtain AD Hessians. Both MLIPs follow the overall DFT trend (left), but their projected forces oscillate relative to DFT (middle). AD amplifies these fluctuations into large Hessian errors along the reaction path (right). Together with the spurious negative modes and underestimated orthogonal curvatures, this suggests that AD Hessian errors may arise from noise in the underlying MLIP potential energy surface.

2.3 Computational efficiency and scaling

The computational cost of Hessian prediction increases with system size. Fig. 3a,b compares the inference time and memory footprint of HIP and AD approaches. HIP has the same asymptotic scaling as a regular forward pass, with a 25% time overhead, approximately the cost of one additional layer on top of the four hidden layers. AD approaches must instead backpropagate through the graph for each row of the Hessian, increasing the cost to approximately $3N$ times that of a forward pass. Quantitatively, HIP yields a 10× to 70× speedup on small molecules (5-30 atoms) and reduces peak memory usage by a factor of 2-3 compared to AD. This efficiency increases the number and size of systems for which Hessian-based workflows are computationally practical. AD calculations are also harder to parallelize across a batch dimension. Because HIP reuses the message-passing machinery, it can be batched like the forward energy pass. At the largest evaluated HIP batch size, its per-sample inference time is at least 81× lower than that of the batch-one AD Hessian implementation of Cui et al. [23].

2.4 Generalization to larger systems

To assess extrapolation beyond the training distribution, we evaluate HIP on HORM-RGD1 and on 710 PubChem samples containing 30–100 atoms (10 samples per atom count) [34], using reference Hessians calculated at matched DFT levels (Methods, Data and DFT calculations). Fig. 3c,d shows the Hessian eigenvalue error as a function of atom count. We use eigenvalues because their scale is independent of system size, whereas the element-wise Hessian MAE can become misleadingly low as an increasing fraction of the elements approaches zero in larger systems. HIP errors remain relatively flat across system sizes, including for molecules larger than those in the training set. In contrast, the errors and computational costs of AD-based predictions increase for larger molecules.

2.5 Downstream utility

2.5.1 Geometry optimization

We evaluate HIP Hessians in geometry-optimization workflows by comparing exact second-order rational-function optimization (RFO) using Hessians with first-order methods and quasi-second-order RFO using the Broyden–Fletcher–Goldfarb–Shanno (BFGS) approximation. Fig. 4a shows convergence statistics across the validation set. RFO with HIP Hessians converges in the fewest steps on average, whereas finite-difference and AD Hessians often fail to converge within a 150-step budget. Naïve steepest descent rarely converges, underscoring the importance of curvature information. The wall-clock timings in Fig. 4a show the same trend. RFO with HIP Hessians is fastest overall because it requires fewer optimization steps and less expensive Hessian evaluations, whereas AD and finite-difference approaches are orders of magnitude slower.

2.5.2 Transition state search

Transition-state searches require accurate Hessian eigenvalues and eigenvectors. We evaluate HIP using the ReactBench benchmark [25] with additional DFT verification (Methods, Transition-state search). The workflow comprises four stages. The growing string method (GSM) initially approaches the TS using energies and forces before RFO optimization with Hessians refines the TS geometry. A DFT Hessian calculation confirms the saddle point, followed by an intrinsic reaction coordinate (IRC) calculation to verify connectivity. Fig. 4b shows the success rates at each stage. GSM performs equally well for both models because it relies only on energies and forces, which are identical. The differences emerge in Hessian-dependent stages. HIP achieves higher RFO convergence, better IRC verification and more DFT-verified transition states. When requiring both DFT verification and tight convergence, HIP improves 12% over AD. These results show that the lower Hessian errors of HIP translate to more reliable TS identification for estimating reaction barriers and rates.

2.5.3 Zero-point energy

Zero-point energy is a thermochemical property derived from vibrational frequencies, which are computed from the Hessian eigenvalues. We evaluate both absolute ZPE at the reactant geometry and relative ΔZPE between reactants and products (Methods, ZPE and extrema classification). Fig. 4c shows lower errors with HIP. For absolute ZPE, the HIP error is one order of magnitude lower than that of LeftNet-DF and two orders of magnitude lower than that of AD-EquiformerV2. For relative ΔZPE, HIP reduces the error by 97.03% compared with AD-EquiformerV2. These results show that the improved eigenvalue predictions of HIP yield more accurate thermochemical properties, where small errors in vibrational frequencies can lead to large errors in free energy.

2.5.4 Frequency analysis for extrema classification

Characterizing stationary points on potential energy surfaces requires determining the number of negative Hessian eigenvalues: zero for minima, one for first-order transition states, and two or more for higher-index saddle points [36]. We test this capability on 1000 geometries from the HORM-T1x validation, comprising 44% first-order TSs, 37% minima, and 19% higher-order saddle points, as determined from DFT Hessians. Fig. 4d shows classification accuracy based on counting negative eigenvalues. HIP achieves 92% accuracy, 10 percentage points better than the next-best AD model and 17 percentage points better than AD-EquiformerV2. This substantial improvement stems from HIP's more accurate eigenvalue predictions, particularly for the small eigenvalues that determine the character of the stationary point. Accurate extrema classification supports automated reaction-pathway discovery and verifies that optimizations have reached the intended type of stationary point.

2.6 Synergistic improvements from Hessian training

Thus far, we trained only the Hessian readout while freezing the backbone, isolating Hessian quality from energy and force predictions. We now investigate joint end-to-end training of the full model, including Hessian supervision. We train both the standard EquiformerV2 (energy and forces only) and HIP-EquiformerV2 (energy, forces and Hessians) for equal wall-clock time across various dataset sizes. Fig. 5a–c shows that Hessian supervision improves Hessian, energy and force accuracy across all data regimes.

On 10,000 validation structures (Fig. 5d-f), force and Hessian errors are positively correlated for Hessian-trained models, but nearly independent for an energy-force-only baseline, which can achieve low force errors while Hessian MAE remains an order of magnitude higher. HIP occupies a tighter, lower-error region than AD, indicating a more self-consistent potential energy surface.

This synergistic effect suggests that Hessian information provides a richer training signal, helping the model learn a better potential energy surface representation.

3 Discussion

The practical implication of HIP is that second-order information can be used more routinely in computational workflows where its cost has previously been prohibitive. More broadly, expensive tensorial quantities with known symmetries may be represented by dedicated equivariant readouts, removing the need for costly differentiation of a learned scalar potential.

Direct prediction, however, does not guarantee that HIP Hessians are globally integrable or exactly consistent with the predicted energies and forces. Unlike non-conservative force errors, which can cause systematic energy drift during long molecular dynamics trajectories, Hessian errors in the applications considered here enter as local curvature estimates. This is consistent with the widespread use of approximate Broyden–Fletcher–Goldfarb–Shanno (BFGS) Hessians, which indicates that exact consistency is not necessary for all workflows. Applications requiring exact energy–force–Hessian consistency may therefore need additional architectural or loss-based constraints.

Other limitations are shared with existing MLIPs. Accuracy depends on the chemical and configurational coverage of the training data, and the finite interaction cutoff may omit long-range curvature contributions in ionic, polar or extended systems. Moreover, generating reference Hessians makes broad training coverage more expensive than for energy- and force-only models.

One possible way to reduce this data cost is to bootstrap HIP with approximate Hessian labels obtained by automatic differentiation of an energy-and-force model, followed by fine-tuning on a smaller set of reference DFT Hessians. The utility of such pseudo-label pre-training will depend on whether fine-tuning can correct systematic errors and noise in the initial AD Hessians, including the curvature errors observed here. Quantifying the trade-off between pseudo-label quality and the amount of reference Hessian data is therefore an important direction for future work.

Future experiments should validate HIP across broader chemical spaces and higher-fidelity electronic-structure methods. For periodic systems, the local construction can be adapted using a periodic neighbour list. The real-space supercell Hessian could then be Fourier transformed into the momentum-dependent dynamical matrix. The same symmetry-based, direct-prediction approach could also be extended to higher-rank response tensors. For example, the elastic tensor would require an additional global rank-4 readout containing $l = 4$ features, while predicting the cubic force constant tensor would require an atom-triplet rank-3 readout containing $l = 1$ and $l = 3$ features.

4 Methods

4.1 Graph representation

Graph neural networks represent a molecule with N atoms as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ in 3D Euclidean space $\mathbb{R}^3$, with nodes/atoms $I \in \mathcal{V}$ at positions $\mathbf{r}_I \in \mathbb{R}^3$ and edges $(I, J) \in \mathcal{E}$ defined by a cutoff radius. At layer t, each node carries a feature $\mathbf{h}_I^{(t)}$ that is a direct sum of irreducible SO(3) representations (irreps): $\mathbf{h}_I^{(t)} = \bigoplus_{l=0}^{l_{\max}} \mathbf{h}_I^{(t,l)}$, where the angular-momentum degree l runs up to $l_{\max}$ and $\mathbf{h}_I^{(t,l)} = \{\mathbf{h}_{I,l,m}^{(t)}\}_{m=-l}^{l}$ has $2l+1$ components indexed by m. To update the node's features, the model sends messages between connected nodes, which are then aggregated in a permutation-invariant manner (typically the sum or an attention-weighted sum) and added to the receiving node. In SO(3)-equivariant GNNs, node features transform under a global rotation $\mathbf{Q}$ via Wigner $\mathbf{D}$-matrices:

$$\mathbf{h}_{I,l,m}^{(t)}\big(\mathbf{Q}\{\mathbf{r}_k\}_{k=1}^{N}\big) = \sum_{m'=-l}^{l} \mathbf{D}_{mm'}^{(l)}(\mathbf{Q})\, \mathbf{h}_{I,l,m'}^{(t)}\big(\{\mathbf{r}_k\}_{k=1}^{N}\big)\,, \tag{1}$$

Translation equivariance is enforced by building messages from relative displacements $\mathbf{r}_{I,J} = \mathbf{r}_J - \mathbf{r}_I$. If we want O(3) instead of SO(3) equivariance, each order-l feature has an extra parity label that determines the transformation under a global coordinate inversion. Two irreps $\mathbf{p}_{l_1}$ and $\mathbf{g}_{l_2}$ interact using the Clebsch-Gordan tensor product [13]

$$\mathbf{h}_{l_3,m_3} = \Big(\mathbf{p}_{l_1,m_1} \otimes_{l_1,l_2}^{l_3} \mathbf{g}_{l_2,m_2}\Big)_{m_3} = \sum_{m_1=-l_1}^{l_1} \sum_{m_2=-l_2}^{l_2} \mathbf{C}_{(l_1,m_1),(l_2,m_2)}^{(l_3,m_3)} \mathbf{p}_{l_1,m_1} \mathbf{g}_{l_2,m_2} \tag{2}$$

where $\mathbf{C}_{(l_1,m_1),(l_2,m_2)}^{(l_3,m_3)}$ are Clebsch–Gordan coefficients coupling degrees l_1 and l_2 into l_3. A message from source atom J to target atom I is then constructed as

$$\mathbf{v}_{I,J,l_3} = \mathbf{v}_{l_3}(\mathbf{h}_I, \mathbf{h}_J, \mathbf{r}_{I,J}) = \sum_{l_1,l_2} w_{l_1,l_2,l_3}(||\mathbf{r}_{I,J}||) \left(\mathbf{f}_{l_1}(\mathbf{h}_I, \mathbf{h}_J) \otimes_{l_1,l_2}^{l_3} \mathbf{Y}_{l_2}\left(\frac{\mathbf{r}_{I,J}}{||\mathbf{r}_{I,J}||}\right)\right) \tag{3}$$

where $\mathbf{Y}_{l_2}$ is the $2l_2+1$ dimensional vector of the spherical harmonics of degree l_2, $w_{l_1,l_2,l_3} : \mathbb{R} \to \mathbb{R}$ is a learned weighting function, and $\mathbf{f}_{l_1}$ a simple function that takes both node features in, for example concatenation in the case of EquiformerV2. A more efficient variation of Eq. (3) relying on the reduction of the $SO(3)$ tensor product to the $SO(2)$ tensor products was proposed by eSCN [37] and adopted by subsequent works, such as EquiformerV2. Finally, the messages are pooled to update the node features, which can be done, for example, via a sum or using graph attention [32].

4.2 Hessian symmetry requirements

Hessians are symmetric, real-valued, Cartesian tensors. This means, under rotation of the coordinate system with rotation matrix $\mathbf{Q}$, each Hessian sub-block $\mathbf{H}_{I,J}$ transforms as

$$\mathbf{H}_{I,J} \xrightarrow{\mathbf{Q}} \mathbf{Q}\mathbf{H}_{I,J}\mathbf{Q}^\top \tag{4}$$

Any physically meaningful Hessian must satisfy this rotation symmetry, as well as the symmetry of the upper and lower triangular $\mathbf{H} = \mathbf{H}^\top$.

4.3 Hessian prediction head

Starting from T layers of message passing in the backbone, we use the atom features $\mathbf{h}^{(T)}_{I,m,c}$, where c indexes feature channels, and feed them into a Hessian readout module. Here, we add another Hessian readout-specific EquiformerV2 layer to improve expressivity. We then construct the Hessian sub-blocks: We send normal messages with Eq. (3), but instead of attention and summing, we treat the messages as atom-pair features:

$$\mathbf{h}_{I,J,l,m,c} = \mathbf{v}_{I,J,l,m,c} = \mathbf{v}\big(\mathbf{h}^{(T)}_I, \mathbf{h}^{(T)}_J, \mathbf{r}_{I,J}\big) \tag{5}$$

As we are reusing the message passing machinery with an interaction cutoff radius, predicting Hessians scales as $O(N^2)$ initially in memory and compute within the cutoff, and then reduces to $O(N)$ scaling for larger systems. This sparsity-with-distance assumption aligns with the sparsity of Hessians due to the locality of electron interactions, which is common in quantum chemistry and can be mathematically rigorously justified [38]. After message passing, we transform the atom pair features $\mathbf{h}_{I,J,l,m,c}$ into the corresponding Hessian sub-blocks $\mathbf{H}_{I,J}$. We project the pair feature irreps down to a single $\{\tilde{\mathbf{h}}_{I,J,l,m}\}_{l\in\{0,1,2\}}$ irreps feature (`1x0e + 1x1e + 1x2e` in `e3nn` notation [39]) using a linear layer $\mathbf{W}_{l,c}$: $\tilde{\mathbf{h}}_{I,J,l,m} = \mathbf{W}_{l,c}\mathbf{h}_{I,J,l,m,c}$. We then expand $\tilde{\mathbf{h}}_{I,J,l,m}$ to an intermediate Hessian sub-block $\mathbf{H}'_{I,J}$ using the tensor product expansion [40]:

$$\mathbf{H}'_{I,J,m_1,m_2} = \sum_{l,m} \mathbf{C}^{l,m}_{l_1=1,m_1,l_2=1,m_2}\tilde{\mathbf{h}}_{I,J,l,m} \tag{6}$$

where $\mathbf{C}^{l,m}_{l_1,m_1,l_2,m_2}$ are the Clebsch-Gordan coefficients ensuring that the relation in Eq. (4) is enforced [40]. Physically, this is a simple change of basis from the coupled to the uncoupled angular momentum basis [41]. $\mathbf{H}'_{I,J,m_1,m_2}$ is a 3×3 block, since m_1 and m_2 run from $-l_1 = -l_2 = -1$ to $l_1 = l_2 = 1$. As $\mathbf{C}^{l,m}_{l_1=1,m_1,l_2=1,m_2} = 0$ for all $l > 2$, $\mathbf{h}_{I,J,l,m,c}$ only needs to contain irreps up to $l = 2$ to build the 3×3 Hessian sub-blocks. Intuitively, Eq. (6) is the inverse of the irreducible tensor decomposition, so it assembles a spherical tensor back from its irreducible components. Conversely, this also means we need all irreps up to $l = 2$, as including only those up to $l = 1$ would not be sufficient to express the 3×3 tensors that decompose into $l = 2$ irreps. Finally, we symmetrize the intermediate Hessian to the final prediction with $\mathbf{H} = \mathbf{H}' + \mathbf{H}'^\top$.

Any equivariant neural network [29] with features of at least $l = 2$ can be turned HIP by equipping it with a Hessian readout head. In this work, we pick the EquiformerV2 architecture as the backbone [32]. Each layer contains a layer norm, message passing with graph attention, another layer norm, and a feed-forward layer.

4.4 Loss function design

To learn the Hessians, one can use standard loss functions like mean absolute error (MAE) or mean squared error (MSE) between predicted and actual Hessian elements:

$$\mathcal{L}_{\text{MAE / MSE}} = \sum_{i,j} |\mathbf{H}_{i,j} - \mathbf{H}^{\text{pred}}_{i,j}|_{\text{MAE / MSE}} \tag{7}$$

However, we are often mainly interested in the smallest eigenvalues and the corresponding eigenvectors [42, 43]. For this reason, we use a loss function that emphasizes the relevant subspace of the Hessian. This loss function is similar to [44], which uses it to emphasize the subspace up to the Lowest Unoccupied Molecular Orbital (LUMO) in Fock matrix prediction. Let $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_{3N}]$ be the matrix with columns made from eigenvectors of the ground truth Hessian $\mathbf{H}$, and $\mathbf{\Lambda}$ the corresponding matrix with eigenvalues on the diagonal and zeros everywhere else. Further, denote $\mathbf{V}_{[:,:k]}$ and $\mathbf{V}_{[:,k:]}$ the matrices sliced to contain all columns up to and including the k^{th} one. We can then define a subspace loss by

$$\mathcal{L}_{\text{sub}} = \sum_{i,j} \left| \mathbf{V}^{\top}_{[:,:k]} \mathbf{H}^{\text{pred}} \mathbf{V}_{[:,:k]} - \mathbf{\Lambda}_{[:,:k]} \right|_{i,j} \tag{8}$$

If $\mathbf{H}^{\text{pred}} = \mathbf{H}$ we have $\mathbf{V}^{\top}\mathbf{H}^{\text{pred}}\mathbf{V} = \mathbf{\Lambda}$, and the loss has therefore the correct minimum. Using $\mathcal{L} = \mathcal{L}_{\text{MAE/MSE}} + \alpha\mathcal{L}_{\text{sub}}$, with subspace weight $\alpha > 0$, lets us emphasize the subspace corresponding to the k lowest eigenvectors and eigenvalues. Since the Hessian always has six redundant translational and rotational degrees of freedom with eigenvalues close to zero, we need to set $k > 6$. In practice, we use $k = 8$ and $\alpha = 1$. Predicting Hessians with HIP using any loss function substantially outperforms AD Hessians. MAE, combined with the subspace loss in Eq. (8), further improves results in transition state search and extrema classification with frequency analysis by a small margin, which is why we use both in the main text experiment.

4.5 Data and DFT calculations

We train and evaluate HIP on the Hessian dataset for Optimizing Reactive MLIP (HORM) [23], comprising 1,725,362 training geometries from Transition1x (T1x) [33], 50,844 T1x validation geometries, and 60,000 HORM-RGD1 geometries used for size-generalization tests. All HORM energy, force, and Hessian labels were computed with GPU4PySCF v1.3.0 [45] using the ωB97X functional [46] and the 6-31G* basis set [47, 48], equivalently denoted 6-31G(d). Molecules are closed-shell neutrals with charge 0 and spin multiplicity 1. The self-consistent field was converged to an energy change below 10^{-10} Hartree and an orbital-gradient norm below 10^{-5}, the

GPU4PySCF v1.3.0 defaults. Analytical Hessians were obtained from the PySCF `Hessian()` implementation.

Additional DFT calculations performed for this work use the same ωB97X/6-31G* level unless noted. PubChem size-extrapolation references were computed with ORCA 6.0 [35] (see Size-generalization benchmarks). Glycine proton-transfer calculations used ORCA 6.1.1 with ωB97X-D3/6-31G(d) (see Glycine proton-transfer case study). ReactBench DFT verification used GPU4PySCF v1.3.0 at ωB97X/6-31G* (see Transition-state search). Accession details are given in the Data availability section.

4.6 Model training and hyperparameters

To isolate Hessian prediction quality from energy and force predictions, we train only the Hessian readout head while keeping the EquiformerV2 backbone, pre-trained on energies, forces, and AD Hessians, frozen. HIP-EquiformerV2 and AD-EquiformerV2 therefore have identical energy and force predictions. All AD-Hessian baselines were trained on the same DFT Hessians.

The EquiformerV2 backbone and optimizer follow the HORM EquiformerV2 configuration, which we restate here. EquiformerV2 uses four layers, 128 sphere channels, 64 attention hidden, alpha, and value channels, four attention heads, 128 feed-forward hidden channels, SiLU activations, 512 Gaussian distance basis functions, and spherical harmonics with $l_{\text{max}} = 4$ and $m_{\text{max}} = 2$. The grid resolution is 18 with 128 sphere samples. The model and Hessian cutoffs are 12 Å. The Hessian head uses three layers for frozen-backbone training and one for end-to-end training. Dropout and drop path are zero. We use the MAE and subspace losses with weights 1, $k = 8$, and $\alpha = 1$. Models are trained for 500k steps with batch size 128 using AdamW, $\beta = (0.9, 0.999)$, AMSGrad, no weight decay, and gradient clipping at norm 0.1. The learning rate is warmed up for 15k steps from 0 to 5×10^{-4}, then follows cosine decay over 500k steps to a minimum of 5×10^{-5}. HIP is trained on one H100 GPU.

For the data-scaling experiment, we train EquiformerV2 on energies and forces and HIP-EquiformerV2 on energies, forces, and Hessians for the same three-day wall-clock budget across each dataset size.

4.7 Evaluation metrics

We assess Hessian quality using the element-wise mean absolute error (MAE), the MAE of all eigenvalues and of the smallest eigenvalue, and the cosine similarity of the smallest eigenvector. We also measure mean inference time and peak memory. For comparisons of vibrational spectra, both reference and predicted Hessians are mass weighted and projected onto the vibrational subspace using Eckart projection before computing the eigenvalue MAE.

4.8 Size-generalization benchmarks

We evaluate size generalization on HORM-RGD1, using the published GPU4PySCF ωB97X/6-31G* labels, and on 710 neutral PubChem molecules containing only C, H, N, and O, with 30–100 atoms and ten molecules per atom-count bin. For each target atom count N, we enumerated stoichiometrically plausible neutral CHNO formulae and queried the PubChem PUG REST `fastformula` endpoint for up to 12 compound identifiers per formula. Candidates were retained if they had formal charge zero, contained only elements with atomic number $Z \in \{1, 6, 7, 8\}$, and had exactly N atoms in the retrieved 3D structure. When no suitable PubChem 3D conformer was available, we generated one from the isomeric SMILES using RDKit with ETKDGv3 followed by MMFF or UFF relaxation. Remaining atom-count bins were filled with deterministic RDKit-generated CHNO conformers.

For the PubChem set, reference energies, Cartesian gradients, and Hessians were computed on the fixed geometries with charge zero and multiplicity one using ORCA 6.0 [35] at ωB97X/6-31G(d). We used separate `! wB97X 6-31G(d) TightSCF Freq` and `! wB97X 6-31G(d) TightSCF EnGrad` calculations for the analytical Hessian and energy/gradient, respectively. The dataset is available at huggingface.co/andreasburger/hip/tree/main/orca_wb97x_631gd_chno_30_100.

Before bin-wise averaging, we identify outliers from the per-molecule Cartesian Hessian MAE,

$$\mathrm{MAE}(\mathbf{H}) = \frac{1}{9N^2} \sum_{i,j=1}^{3N} \left| H_{ij}^{\mathrm{pred}} - H_{ij}^{\mathrm{ref}} \right|.$$

Let x_i denote this per-molecule MAE. For each atom-count bin, we compute the median $\tilde{x}_N$, median absolute deviation $\mathrm{MAD}_N = \mathrm{median}_i |x_i - \tilde{x}_N|$, and modified z-score $M_i = 0.6745(x_i - \tilde{x}_N)/\mathrm{MAD}_N$ [49]. Outliers with $|M_i| > 10$ are detected independently for each method, and the union is removed from all methods to maintain comparable samples. This removes 10 of 710 PubChem molecules. We apply the same procedure to HORM-RGD1, detecting outliers independently for HIP, AD, and EquiformerV2 trained on energies and forces only (E-F), and remove 72 of 60,000 molecules.

4.9 Geometry optimization

We compare RFO using predicted Hessians and RFO with BFGS updates initialized by a predicted Hessian against steepest descent, FIRE [50], RFO-BFGS initialized from the identity, and RFO using AD or finite-difference Hessians. We relax 80 reactant geometries from the Transition1x validation set after adding noise with 0.5 Å RMS. We use the RS-RFO and redundant-coordinate implementation in pysisyphus [51, 52], Gaussian default convergence criteria, and a budget of 150 steps. The maximum force, RMS force, maximum step, and RMS step thresholds are 4.5×10^{-4}, 3.0×10^{-4}, 1.8×10^{-3}, and 1.2×10^{-3}, respectively.

4.10 Transition-state search

We use ReactBench [25], with optimization routines implemented in pysisyphus [51] and PyGSM [53]. An initial guess is generated using the growing string method [54], followed by local search using RS-P-RFO [42], frequency analysis, and the intrinsic reaction coordinate (IRC). We report success at each stage using the following criteria. "GSM Success" requires a force RMS below 5×10^{-5} Hartree/Bohr within 100 iterations. "TS Success" requires frequency analysis after RS-P-RFO to find exactly one negative Hessian eigenvalue. "IRC Verified" requires the IRC to converge to the same force-RMS criterion and recover the original reactant and product. Samples passing all three stages are treated as transition-state proposals. For a subset of 100 proposals, we recompute single-point DFT energies, forces, and analytical Hessians with GPU4PySCF v1.3.0 [45] at ωB97X/6-31G*, and require the DFT Hessian to have one negative eigenvalue and the DFT force RMS to lie below 2×10^{-3} Hartree/Bohr.

4.11 ZPE and extrema classification

Starting from 80 reactant and product geometries from the Transition1x validation set, we relax the geometries with the RS-RFO implementation in pysisyphus [51], using BFGS updates and identity Hessian initialization, on the MLIP trained at ωB97X/6-31G*. We use Gaussian tight convergence, with maximum force, RMS force, maximum step, and RMS step thresholds of 1.5×10^{-5}, 1.0×10^{-5}, 6.0×10^{-5}, and 4.0×10^{-5}, respectively. We compute ZPE from the Eckart-projected, mass-weighted Hessian, $\tilde{H}_{ij} = H_{ij}/\sqrt{m_i m_j}$, and report energies in eV per molecule. Absolute ZPE error is evaluated at the reactant geometry. For method m, $\Delta\mathrm{ZPE}_m = \mathrm{ZPE}_m(\mathrm{product}) - \mathrm{ZPE}_m(\mathrm{reactant})$, and the model error is $|\Delta\mathrm{ZPE}_{\mathrm{model}} - \Delta\mathrm{ZPE}_{\mathrm{DFT}}|$, where the DFT reference is the ωB97X/6-31G* Hessian at the same geometry.

For extrema classification, we count the negative eigenvalues of the projected Hessian for 1,000 HORM-T1x validation geometries. The DFT labels comprise 44% first-order transition states, 37% minima, and 19% higher-order saddle points.

4.12 Glycine proton-transfer case study

We selected the intramolecular proton transfer of glycine from the Transition1x test split (sample ID 5, reaction 1961). The transferring proton is described by $q_{\mathrm{NH}} = d(\mathrm{N4}, \mathrm{H9})$ and $q_{\mathrm{OH}} = d(\mathrm{O3}, \mathrm{H9})$. For each point on the two-dimensional surface, we started from the Transition1x transition-state structure and placed the proton at the target distances. We excluded pairs violating the triangle inequalities $q_{\mathrm{NH}} + q_{\mathrm{OH}} < d(\mathrm{N4}, \mathrm{O3})$ or $|q_{\mathrm{NH}} - q_{\mathrm{OH}}| > d(\mathrm{N4}, \mathrm{O3})$. All other degrees of freedom were relaxed while both distances remained constrained. Structures were initially relaxed with GFN2-xTB using TBLite and ASE BFGS and then reoptimized with the same constraints using ORCA 6.1.1 [35] and `! wB97X-D3 6-31G(d) TightSCF Opt`. This produced 579 DFT-relaxed geometries. Energies and forces were evaluated with `! wB97X-D3 6-31G(d) TightSCF EnGrad`, and Hessians with `! wB97X-D3 6-31G(d) TightSCF Freq`. The data are available at huggingface.co/andreasburger/hip/tree/main/orca_wb97x_d3_631gd_glycine_pt_dft_relaxed_579.

For the one-dimensional minimum-energy path, we use the final saved block of eight Transition1x NEB images together with the reactant and product. Geodesic interpolation in redundant internal-coordinate space produces 145 fixed path geometries. Single-point energies and forces were computed with ORCA 6.1.1 using `! wB97X-D3 6-31G(d) TightSCF EnGrad`, and analytical Hessians with `! wB97X-D3 6-31G(d) TightSCF Freq`. The data are available at huggingface.co/andreasburger/hip/tree/main/orca_wb97x_631gd_glycine_pt_mep_145.

5 Data availability

Source data for Figures 2–5 is available with this manuscript. The HORM training and validation dataset used in this study is available from Zenodo at https://doi.org/10.5281/zenodo.17217897 [23, 55]. HORM geometries originate from Transition1x [33], available from figshare at https://doi.org/10.6084/m9.figshare.19614657.v4 [56]. DFT calculations generated in this work, including atomic coordinates, energies, forces and Hessians for the PubChem CHNO size-extrapolation set, the glycine proton-transfer surface and the glycine minimum-energy path, together with trained HIP model weights, are available from Zenodo at https://doi.org/10.5281/zenodo.22003643 [57] and from Hugging Face at https://huggingface.co/andreasburger/hip.

6 Code availability

The HIP source code for training, inference and evaluation is available at https://github.com/BurgerAndreas/hip under the Apache License, Version 2.0. A frozen snapshot is archived at Zenodo at https://doi.org/10.5281/zenodo.22003592 [58]. All requests should be addressed to Andreas Burger (me@andreas-burger.com).

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9 Author contributions

Code, training, and experiments by A.Bu.; idea, theory, and Clebsch–Gordan transform code by L.T.; paper writing by L.T., A.Bu. and N.R.; visualizations by A.Bu. and L.T.; supervision and paper editing by A.Bh., T.V., V.B. and A.A.-G.; manuscript review and approval by A.Bu., L.T., N.R., N.V., V.B., T.V., A.Bh. and A.A.-G.

10 Competing interests

The authors declare no competing interests.

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Tables

Model	Hessian Method	Hessian Trained	$H \downarrow$ eV/Å^2	$\lambda \downarrow$ eV/Å^2	Cos $\boldsymbol{v}_1 \uparrow$ unitless	$\lambda_1 \downarrow$ eV/Å^2	Time $\downarrow$ ms
AlphaNet (E-F)	AD		0.502	1.190	0.903	0.245	728.6
LEFTNet-DF (E-F)			1.650	2.247	0.505	1.362	331.4
LEFTNet-CF (E-F)			0.364	1.011	0.947	0.130	1047.9
EquiformerV2 (E-F)			2.254	4.199	0.279	1.372	564.9
AlphaNet	AD	✓	0.390	0.790	0.899	0.244	747.0
LEFTNet-DF		✓	0.200	0.172	0.937	0.136	332.2
LEFTNet-CF		✓	0.153	0.226	0.951	0.127	1051.6
EquiformerV2		✓	0.077	0.070	0.916	0.104	562.3
HIP-EquiformerV2	HIP	✓	0.030	0.063	**0.982**	**0.031**	38.5
HIP-EquiformerV2*		✓	**0.020**	**0.041**	**0.982**	**0.031**	**31.4**

Table 1 Hessian prediction errors on the HORM-T1x validation set. Denoted are the mean absolute errors (MAE), cosine similarity (Cos), and average time per forward pass on an RTX 3060 GPU. λ_1 and $\boldsymbol{v}_1$ were obtained after Eckart-projection. Bold values highlight the best-performing model. Baseline AD models are taken from [23]. In HIP-EquiformerV2 we trained only the Hessian readout head while keeping the parameters of the backbone frozen, while HIP-EquiformerV2* is trained end-to-end.

Figure legends

Fig. 1 Hessian interatomic potential architecture. A conventional machine-learning interatomic potential (MLIP; grey) maps a molecular geometry to its energy and forces. A Hessian interatomic potential (HIP) augments the MLIP with an equivariant Hessian readout that combines node contributions (orange) and pairwise message contributions (green) to predict each Hessian block, $H_{ij} = \partial^2 E/\partial \boldsymbol{R}_i \partial \boldsymbol{R}_j$, directly. Conventional MLIPs instead construct the Hessian by costly automatic differentiation (AD) of the predicted energy or forces. The predicted Hessian supports geometry optimization, transition-state searches, frequency analysis and zero-point-energy calculations (green). HIP can use both direct force prediction and conservative forces.

Fig. 2 Glycine proton transfer. (a) Number of negative eigenvalues of the Eckart-projected, mass-weighted Hessian over the two-dimensional surface defined by the N–H and O–H distances, q_{NH} and q_{OH}. Density-functional theory (DFT) provides the reference. Hessian interatomic potential (HIP) denotes direct Hessian prediction and automatic differentiation (AD) denotes Hessians differentiated from learned forces. Shading gives the integer negative-mode count and black lines are DFT energy contours. Circles, stars and triangles mark the reactant, transition state and product, respectively. In the molecular rendering, carbon is grey, hydrogen white, nitrogen blue and oxygen red. **(b)** Six lowest Hessian eigenvalues, λ_1–λ_6, along 145 minimum-energy-path (MEP) geometries versus the collective variable $\xi = q_{\mathrm{NH}} - q_{\mathrm{OH}}$. DFT is shown as grey points, HIP as solid orange lines and EquiformerV2 (EqV2) AD as dotted blue lines. **(c)** Force projected onto the unit tangent $\widehat{\boldsymbol{\xi}}$ of the MEP (left), force residual relative to DFT (middle) and Hessian mean absolute error (MAE; right). DFT is shown as grey points, EqV2 AD as dotted blue lines and LeftNet-CF AD as solid purple lines.

Fig. 3 Computational cost and molecular-size scaling. Hessian interatomic potential (HIP) denotes direct Hessian prediction, automatic differentiation (AD) denotes Hessians differentiated from learned forces or energies, and finite difference (FD) denotes numerical force differentiation. **(a,b)** Mean single-molecule inference time (a) and peak graphics-processing-unit (GPU) memory (b) versus atom count on an RTX 3060 GPU; shading shows one standard deviation over replicates. The forward pass is teal and dotted, HIP orange, direct-force AD blue, conservative-force AD purple and FD dark grey. The forward-pass and HIP curves overlap. **(c,d)** Mean eigenvalue mean absolute error (MAE) of the Eckart-projected Hessian versus atom count for HORM-RGD1 (c) and ten PubChem samples per atom count (d) [34]; shading shows one standard deviation over samples. HIP is shown by orange diamonds and AD by blue circles. The vertical grey dashed line in panel c marks the largest molecule size represented during training.

Fig. 4 Molecular-structure applications. Hessian interatomic potential (HIP) denotes direct Hessian prediction, automatic differentiation (AD) denotes Hessians differentiated from learned forces and density-functional theory (DFT) provides the reference. Colours distinguish methods and correspond to the adjacent axis labels. **(a)** Optimization steps and wall-clock time for 80 geometries. Violin shading represents the sample distribution, overlaid points represent individual geometries and crosses denote calculations that did not converge within 150 steps. Rational-function optimization (RFO) wall-clock times with AD or finite-difference Hessians are omitted because they are one to two orders of magnitude longer than RFO with HIP. **(b)** Successful ReactBench samples after the growing string method (GSM), RFO convergence, intrinsic reaction coordinate (IRC), DFT transition-state (TS) verification and an additional tight force-convergence criterion, from 960 initial reactions [25]. DFT-verified full-set counts are extrapolated from fixed subsets of 100 TS proposals per Hessian method. **(c)** Mean absolute reactant zero-point-energy (ZPE) error and reaction ΔZPE error relative to DFT over 80 reactant–product pairs; the vertical axis is logarithmic. **(d)** Fraction of 1,000 stationary points for which the exact number of negative Hessian eigenvalues is predicted. Error bars in panels c and d show 95% bootstrap confidence intervals.

Fig. 5 Dataset scaling and force–Hessian error correlation. Mean absolute error (MAE) is evaluated on the HORM-T1x validation set. **(a–c)** Energy (a), force (b) and Hessian (c) MAE versus the number of randomly selected training samples. EquiformerV2 trained on energies and forces is shown in dark grey; the Hessian interatomic potential (HIP) trained end-to-end on energies, forces and Hessians is shown in orange. **(d–f)** Per-geometry force and Hessian MAE for 10,000 validation geometries using EquiformerV2 with Hessian supervision and Hessians obtained by automatic differentiation (AD; d), HIP trained end-to-end (e) and EquiformerV2 trained only on energies and forces with Hessians obtained by AD (f). Each point represents one geometry. The colour scale, from dark purple to yellow, indicates atom count. The displayed log-r is the Pearson correlation coefficient between the base-10 logarithms of the force and Hessian errors.